Synopsis

Disease Prediction using Machine Learning

Project ID : PY-GT-48

Problem ID : 1

Student Name : Lavya Rautela

University Roll number : 2028754

Section : G

Student Name : Snehil Joshi

University Roll number : 2028861

Section : G

Student Name : Suryansh Tayal

University Roll number : 2028874

Section : G

- ## Problem Description:

Machine learning is widely used in healthcare to analyze patient data and assist in early disease prediction. Many diseases such as heart disease can be predicted by studying clinical parameters like age, blood pressure, cholesterol level, blood sugar, heart rate, and other medical attributes. Manual analysis of large patient records is difficult and time-consuming, so an automated system can be very useful.

The objective of this project is to build a simple machine learning classification system that predicts whether a patient is at risk of disease or not using clinical data. The project uses a healthcare dataset and applies preprocessing techniques such as data cleaning and normalization to improve data quality. Three machine learning algorithms—Logistic Regression, K-Nearest Neighbours (KNN), and Support Vector Machine (SVM)—are used for training and testing.

The models are evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. This project demonstrates a complete beginner-level machine learning pipeline and shows how AI can support decision-making in the healthcare field.

- ## Steps of Implementation:

This project follows a structured machine learning pipeline for disease prediction in healthcare. Each phase of implementation is explained below:

1. User Authentication

A basic interface or system access control is implemented to ensure only authorized users (e.g., healthcare professionals, researchers) can access the disease prediction system. This maintains patient data security and controlled usage.

2. Record Handling

The healthcare dataset is collected and managed efficiently. Records containing features like age, blood pressure, cholesterol level, blood sugar, heart rate, and other clinical parameters are stored, updated, and organized for further processing.

3. String Manipulation

Any textual or categorical data (such as patient IDs, gender, or disease labels) is cleaned and formatted. Labels like "Disease" and "No Disease" are standardized and converted into machine-readable format.

4. File Handling

Dataset files (CSV/Excel) are read, written, and managed using file handling techniques. Cleaned and processed data is stored for reuse, and model outputs/results are saved for reporting.

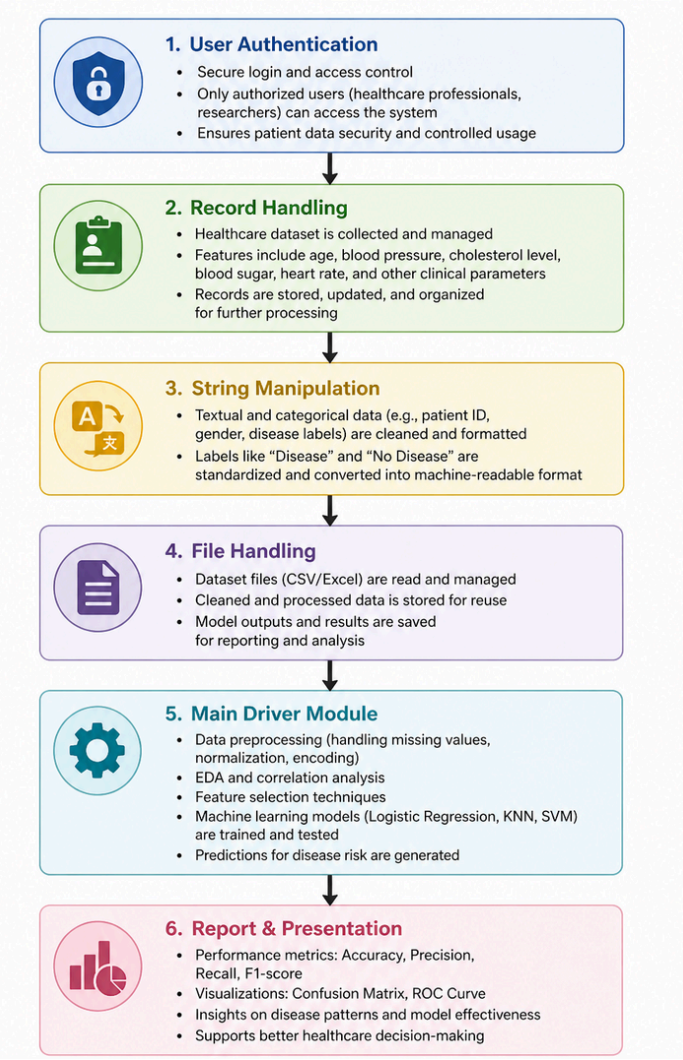
5. Main Driver Module

This is the core part of the system where:

- Data preprocessing is performed
- EDA and correlation analysis are conducted
- Feature selection techniques are applied
- Machine learning models (Logistic Regression, KNN, SVM) are trained and tested
- Predictions for disease risk are generated

6. Report & Presentation

The final results are presented using performance metrics (accuracy, precision, recall, F1-score) and visualizations including confusion matrix and ROC curve. Insights such as disease patterns and model effectiveness are summarized to help in better healthcare decision-making.


1. User Authentication
Secure login and access control
Only authorized users (healthcare professionals, researchers) can access the system
Ensures patient data security and controlled usage
2. Record Handling
Healthcare dataset is collected and managed
Features include age, blood pressure, cholesterol level, blood sugar, heart rate, and other clinical parameters
Records are stored, updated, and organized for further processing
3. String Manipulation
Textual and categorical data (e.g., patient ID, gender, disease labels) are cleaned and formatted
Labels like "Disease" and "No Disease" are standardized and converted into machine-readable format
4. File Handling
Dataset files (CSV/Excel) are read and managed
Cleaned and processed data is stored for reuse
Model outputs and results are saved for reporting and analysis
5. Main Driver Module
Data preprocessing (handling missing values, normalization, encoding)
EDA and correlation analysis
Feature selection techniques
Machine learning models (Logistic Regression, KNN, SVM) are trained and tested
Predictions for disease risk are generated
6. Report & Presentation
Performance metrics: Accuracy, Precision, Recall, F1-score
Visualizations: Confusion Matrix, ROC Curve
Insights on disease patterns and model effectiveness
Supports better healthcare decision-making

- ## Proposed Modules:

1. Data Collection Module

Handles the selection and loading of the healthcare dataset containing clinical features like age, blood pressure, cholesterol level, blood sugar, heart rate, and other medical attributes.

2. Data Preprocessing Module

Cleans the dataset by handling missing values, removing duplicates, and applying normalization/standardization to ensure consistency and quality of medical data.

3. Exploratory Data Analysis (EDA) Module

Performs statistical analysis and visualizations (histograms, scatter plots, box plots, heatmaps) to understand data patterns, detect anomalies, and identify disease risk trends.

4. Feature Selection Module

Selects the most relevant clinical features using statistical techniques to improve model accuracy and reduce redundancy in medical parameters.

5. Model Training Module

Implements machine learning algorithms such as Logistic Regression, K-Nearest Neighbours (KNN), and Support Vector Machine (SVM) to train models for disease prediction.

6. Model Evaluation Module

Evaluates model performance using metrics like accuracy, precision, recall, F1-score, confusion matrix, and ROC curve to select the best model for healthcare predictions.

7. Prediction Module

Uses the trained model to predict disease risk based on new patient clinical data inputs.

8. Reporting & Visualization Module

Generates reports and visual outputs including ROC curves and confusion matrices to present insights, model results, and disease patterns for better decision-making by healthcare professionals.

- ## Required topics from the subject:

1. File Handling

File handling is used to read and write the healthcare dataset stored in CSV/Excel format. It helps in loading raw patient data, saving cleaned data, and storing intermediate results during preprocessing and EDA.

2. Data Structures (Arrays/DataFrames)

Data structures like arrays and DataFrames (in Python) are used to store and manipulate large amounts of patient data efficiently. They allow easy access, modification, and analysis of clinical features such as blood pressure and cholesterol levels.

3. Data Preprocessing Techniques

Includes handling missing values, removing duplicates, and scaling data using normalization or standardization. These techniques improve data quality and ensure better performance of medical data analysis.

4. Exploratory Data Analysis (EDA)

EDA involves analyzing patient data using statistical summaries and visualizations like histograms, scatter plots, and box plots. It helps in understanding patterns, detecting anomalies, and identifying disease risk trends.

5. Basic Statistics

Statistical concepts like mean, median, standard deviation, and correlation are used to summarize and understand the healthcare dataset. They help in identifying relationships between clinical parameters and disease risk.

6. Visualization Techniques

Graphs and plots are used to represent medical data visually for better understanding. Tools like matplotlib and seaborn help in creating heatmaps, histograms, scatter plots, confusion matrices, and ROC curves during analysis.

- ## Platform Required:

Jupyter Notebook is used as the primary platform for implementing the project. It provides an interactive environment for writing and executing Python code, performing data preprocessing, and conducting Exploratory Data Analysis (EDA). It also supports visualization libraries like matplotlib and seaborn, making it suitable for analyzing and presenting healthcare data effectively, including generating confusion matrices and ROC curves for model evaluation.

- ## Books and Link Sources:

1. Pandas Documentation (Data Handling & Preprocessing)
https://pandas.pydata.org/docs/
Used for reading healthcare datasets, handling missing values, and performing data preprocessing on patient records.

2. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow
By - Aurélien Géron (Book)
A comprehensive book covering the complete ML pipeline including preprocessing, EDA, feature selection, model building, and evaluation. Useful for understanding end-to-end healthcare prediction system design.

3. Scikit-learn Official Documentation (ML Models & Preprocessing)
https://scikit-learn.org/stable/
Covers data preprocessing, feature selection methods, and classification algorithms like Logistic Regression, SVM, KNN, etc. Also includes documentation for confusion matrix and ROC curve generation.

4. Kaggle Learn (Practical ML & EDA Tutorials)
https://www.kaggle.com/learn
Offers hands-on tutorials on data preprocessing, EDA, feature engineering, and machine learning workflows specifically applicable to healthcare datasets.

5. Healthcare Machine Learning Research Papers (IEEE Xplore / Google Scholar)
https://scholar.google.com/
Used to study real-world applications of disease prediction and understand how ML is applied in healthcare analysis and clinical decision support systems.